MILITARY SPECIFICATION

KNOBS, CONTROL (FOR USE WITH ELECTRONIC, COMMUNICATIONS, AND ALLIED EQUIPMENT).

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification covers the general requirements for manually operated control knobs and associated knob locks for use with electronic, communications, and allied equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-S-200	- Setscrews; Hexagon Socket and Spline Socket, Headless.
TT-P-54	- Paint, Phosphorescent, Ready-mixed (non-radioactive).
GGG-K-275	- Key, Socket Head Screw, (Wrench, Hexagon, and Spline)
PPP-T-60	- Tape; Pressure Sensitive Adhesive, Waterproof for Packaging and Sealing.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-3891	- Luminescent Material and Equipment (non-radioactive).
MIL-F-7179	- Finishes and Coatings, General Specification for Protection of Aircraft and Aircraft Parts.
MIL-P-7788	- Plate, Plastic, Cockpit and Interior Controls Lighting.
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-C-18012	- Control Configuration and Markings; For Plastic Lighting Plates, Control Panels and Placards.
MIL-C-45662	- Calibration System Requirements.

STANDARDS

FEDERAL

FED-STD-595 - Colors

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105	- Sampling procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	- Military Standard Environmental Test Methods for Aerospace and Ground Equipment.
MIL-STD-456	- Electronic Parts, Date and Source Coding for.

(For applicable MS Military Standards, see Supplement 1).

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Military standards for individual knob types. - All requirements specified herein apply to all types of knobs covered by applicable MS military standards. Details or exceptions applicable to individual types are specified on the MS military standards listed on Supplement 1 to this specification. In the event of any conflict between this specification and the individual MS military standards, the latter shall govern. (See 6.1)

3.2 Qualification. - Knobs furnished under this specification shall be a product which has been tested and has passed qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.3 Material. - The material for each part shall be as specified. (See 3.1). However, when a definite material is not specified, a material shall be used which will enable the knobs to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Fluorescent. - Unless otherwise specified (See 3.1), fluorescent material used for calibration marking for black lighted conditions shall conform to type F, green of MIL-L-3891.

3.3.2 Phosphorescent. - Unless otherwise specified (See 3.1), phosphorescent material used for calibration marking shall conform to Type P, green to yellowish green of MIL-L-3891 or type I of TT-P-54.

3.4 Design and construction. -

3.4.1 Form and dimensions. - Knobs shall be made in the form and within the dimensions specified (see 3.1).

3.4.1.1 Concentricity and perpendicularity. - The periphery of the knob shall be concentric to the rotation axis within 0.025-inch total indicator reading (TIR). The back plane of the knob shall be perpendicular within ±3° to the rotational axis of the knob.

3.4.2 Insert. - The insert shall be a non-rotatable, metallic, full cylinder, completely forming the shaft hole. The insert shall be molded in the center of the knob body. It shall be provided with two UNC-3B threaded holes for setscrews as specified in 3.4.3. A 45 degree chamfer of 1/32 ± 1/64 inch shall be provided in the insert end which admits the shaft. A reducing bushing may be pressed into the molded-in bushing as required for 1/8 inch shaft hole types and special configurations.

3.4.3 Setscrews. - Unless otherwise specified (see 3.1), setscrews shall conform to type I, style 4 of FF-S-200. Two No. 4-40 UNC-3A setscrews shall be provided in all 1/8 inch shaft hole diameter knobs and for knobs having a base diameter of 3/4 inch or less. Two No. 8-32 UNC-3A setscrews shall be provided in all other knobs. Two setscrews shall be used in each knob, and unless otherwise specified (see 3.1), they shall be placed between nominal limits of 90 and 125 degrees apart. When an insert for a D flat shaft is provided unless otherwise specified (see 3.1), one of the setscrews shall be perpendicular ± 3 degrees to the flat portion of the insert and opposite any indicator mark on the knob or dial (see 3.4.6). The length of each sets screws shall be such that when the setscrew is set firmly against the shaft, its head shall not be less than

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

0.015 inch and not more than 0.240 inch below the external surface of the knob adjacent to the setscrew hole. The setscrew holes within the plastic knob body up to the metal insert must be counterbored to the minimum major diameter of the setscrew.

3.4.4 Color.- The color of the knobs shall be as specified (see 3.1) in accordance with FED-STD-595. Letters, numbers, or calibration markings shall be as specified for color (see 3.1) in accordance with FED-STD-595. Except in the case of letters, numbers, calibration markings, and trans-illuminated knobs, the color of plastic knobs shall not be applied by coating or painting but shall be incorporated in the material. At the option of the supplier, the finish may be obtained by means of a secondary operation.

3.4.5 Transillumination.- Transilluminated part of the knobs shall meet the requirements of MIL-P-7788 except for materials, thickness, and back coating. Color shall be as specified (see 3.1 and 3.4.4). Letters, numbers, and calibration markings shall be in accordance with MIL-C-18012 or as specified (see 3.1).

3.4.6 Indicator mark.- If the cap or the top part of the knob has a configuration which might be construed as a pointer or indicator, any required orientation or calibration mark, such as an arrow, line, or dot shall be in line with or on such configuration. Unless otherwise specified (see 3.1) the indicator mark shall be located within ± 3 degrees of the center line axis of the setscrews that is diametrically opposite the mark and/or within ± 3 degrees of a line perpendicular to the flat portion of an insert for a "D" flat shaft.

3.5 Resistance to environmental conditions.- When knobs are conditioned as specified in 4.7.3 through 4.7.6 inclusive, there shall be no visible deterioration during and after completion of conditioning. Any of the following shall be considered as deterioration:

(a) Cracking.
(b) Chipping.
(c) Spalling.
(d) Crazing.
(e) Peeling.
(f) Illegibility of Marking. (Marking indicates the identification markings (see 3.10) and calibration or indicator marking by various methods of engraving and filling or hot die stamping. In either case, if after testing the marked characters or figures are discernible the sample shall be acceptable.)
(g) Change in color or luster beyond the specified limits.
(h) Destructive corrosion. (Any type of corrosion which will prevent the setscrews from being turned with the customary amount of torque, or which will not allow the knob to be readily removed from or inserted onto the shaft).

3.6 Torque, tensile, and bending moment tests.-

3.6.1 Setscrews.- When knobs are tested as specified in 4.7.2.1, 4.7.3.2.1, table V, and figure 1, there shall be no failure of the thread in the insert of the knob, setscrew threads, or setscrew socket heads. Insert or setscrew threads and sockets that are stripped, broken, or deformed shall be considered as failure of the thread.

3.6.2 Insert.- When knobs are tested as specified in 4.7.2.2 and 4.7.3.2.2, there shall be no movement of the insert, cracking, or other failure.

3.6.3 Spinner and tactile spinner.- When spinner-type knobs are tested as specified in 4.7.2.3 and 4.7.3, there shall be no evidence of cracking, or of movement of the insert or any other member within the knob, or other failure. Spinner shall turn freely in the spinner knob with no evidence of binding so as to prevent its rotation when the knob is held stationary.

3.6.4 Spinner-knob balance.- When spinner-type knobs are tested as specified in 4.7.2.4, the maximum torque shall not exceed 1 ounce inch.

3.6.5 Caps.- When knobs with caps attached and not molded integrally with the knob body are tested as specified in 4.7.2.5 and 4.7.3.2.2, there shall be no evidence of cracking, looseness, movement or separation of the cap in relation to the body.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

3.6.6 Tactile caps.- When knobs with tactile caps are tested in accordance with 4.7.2.6 and 4.7.3.2.3, there shall be no evidence of cracking, looseness, movement or separation of the cap in relation to the body.

3.6.7 Dial skirts or plates.- When knobs with dial skirts or plates that are attached to the body of a knob are tested as specified in 4.7.2.7, table VI, and 4.7.3.2.4, there shall be no evidence of cracking, looseness, movement, or separation of the skirt in relation to the body.

3.7 Finish.- When knobs are tested as specified in 4.7.4, there shall be no peeling of the lusterless-producing finish. (See 3.1, 3.4.4, and 3.4.5).

3.8 Flammability.- When knobs are tested as specified in 4.7.7, they shall not support visible flame or combustion for more than 30 seconds after removal of the flame.

3.10 Marking.- The following information shall be marked on the knobs in accordance with MIL-STD-130:

a. MS part number.
b. Supplier's name or code symbol and date code in accordance with MIL-STD-456.
c. The MS part number, the supplier's name or code symbol shall be sunken or raised characters, or stamped in contrasting color ink and shall be placed on a flat surface in a position not noticeable to the operator under normal operating conditions. When space does not permit the marking as specified above, the marking shall be applied or stamped on tape or label affixed to the knob.

3.11 Workmanship.- Knobs shall be processed in such a manner as to be uniform in quality and shall be free from pits, cracks, chips, and other defects that will affect serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities.- Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performances of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of the inspection equipment shall comply with the requirements of MIL-C-45662.

4.2 Classification of inspection.- The examination and testing of knobs shall be classified as follows:

(a) Component-materials inspection. (See 4.3).
(b) Qualification inspection (See 4.5).
(c) Quality conformance inspection. (See 4.6).

1. Inspection of product for delivery. (See 4.6.1).
2. Inspection of preparation for delivery. (See 4.6.2).

4.3 Component-materials inspection.- Components-materials inspection shall consist of verification that the component-materials listed in table I, used in fabricating the knobs, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

Table I - Component-materials inspection

Component Material	Requirement Paragraph	Applicable specification or test paragraph
Fluorescent	3.3.1	MIL-L-3891
Phosphorescent	3.3.2	MIL-L-3891 and TT-P-54
Setscrews	3.4.3	FF-S-200

4.4 Inspection conditions.- Unless otherwise specified herein, all inspection shall be made at room ambient temperature, relative humidity, and pressure, in accordance with the general requirements of paragraph 2.2 of MIL-STD-202.

4.5 Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Government. Application for qualification tests shall be made in accordance with "Provisions Governing qualifications SD-6" (See 6.3 and 6.4). Qualification tests shall consist of the tests specified in table II.

4.5.1 Sample units.- The specified number of sample units shall be in accordance with table II for each test for qualification inspection or as applicable (See 4.5.3.2).

4.5.2 Test routine.- Sample units of knobs shall be subjected to the qualification inspection specified in table II in the order shown. All sample units which are to undergo qualification testing shall be subjected to the inspection of group I. The sample units shall then be divided as specified in table II and subjected to the inspection for their particular group.

4.5.3 Extent of qualification.-

4.5.3.1 Single submission.- Qualification shall be restricted to the type submitted.

4.5.3.2 Group submission.- The extent of qualification shall be in accordance with table II and the applicable specification sheet (See 3.1).

Table II - Qualification Inspection

Examination or Test	Requirement Paragraph	Method Paragraph
Group I (All sample units)		
Visual and mechanical examination	3.1, 3.4 to 3.6 Incl 3.10 and 3.11	4.7.1
Group II (3 sample units from group I		
Setscrews	3.6.1	4.7.2.1
Insert	3.6.2	4.7.2.2
Spinner and Tactile spinner	3.6.3	4.7.2.3
Spinner Knob balance	3.6.4	4.7.2.4
Caps	3.6.5	4.7.2.5
Tactile caps	3.6.6	4.7.2.6
Dial skirts or plates	3.6.7	4.7.2.7
Group III (3 sample units from group I)		
Thermal shock	3.5	4.7.3
Moisture Resistance	3.5	4.7.4
Finish	3.7	4.7.5.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

Table II - Qualification Inspection (Continued)

Examination or test	Requirement Paragraph	Method Paragraph
Group IV (3 sample units from group I)		
Sunshine	3.5	4.7.5
Salt spray (corrosion)	3.5	4.7.6
Group V		
Flammability	3.8	4.7.7

4.6 Quality conformance inspection. -

4.6.1 Inspection of product for delivery. - Inspection of product for delivery shall consist of groups A, B, and C.

4.6.1.1 Inspection lot. - An inspection lot, as far as practicable, shall consist of all knobs of the same MS part number, produced under essentially the same conditions.

4.6.1.2 Rejected lots. - If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units, as applicable and reinspect. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots. Rejected lots shall be inspected, using tightened inspection.

4.6.1.3 Group A inspection. - Group A inspection shall consist of the visual and mechanical examinations specified in 4.7.1.

4.6.1.3.1 Sampling plan. - Statistical sampling and inspection shall be in accordance with Standard MIL-STD-105 for ordinary inspection. The acceptable quality levels (AQL) shall be 1.0 and 4.0 percent defective for major and minor defects, respectively. Major defects shall be design, dimension or form (See 4.7.1).

4.6.1.4 Group B inspection. - Group B inspection shall consist of the tests specified in table III, in the order shown. Sample units which have passed group A inspection shall be subjected to group B inspection.

Table III - Group B inspection

Examination or Test	Requirement Paragraph	Method Paragraph
Subgroup I (All lots)		
Torque (at room ambient	3.5 & 3.6	4.7.2
Setscrew	3.6	4.7.2.1
Subgroup II (All lots-spinner types)		
Torque (at room ambient)	3.5 & 3.6	4.7.2
Insert	3.6.2	4.7.2.2
Spinner and tactile spinner	3.6.3	4.7.2.3
Subgroup III (All lots except spinner types and tactiles		
Torque and tensile (at room ambient)	3.5 & 3.6	4.7.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

Table III - Group B inspection (Continued)

Examination or Test	Requirement Paragraph	Method Paragraph
Subgroup III (All lots except spinner types and tactiles)—Cont'd		
Insert	3.6.2	4.7.2.2
Caps	3.6.5	4.7.2.5
Tensile (at room ambient)	3.5 and 3.6	4.7.2
Dial skirts or plates	3.6.7	4.7.2.7
Subgroup IV (All lots with round tactiles)		
Torque and bending moment (at room ambient)	3.5 and 3.6	4.7.2
Insert	3.6.2	4.7.2.2
Tactile caps	3.6.6	4.7.2.6
Tensile (at room ambient)	3.5 and 3.6	4.7.2
Dial skirts or plates	3.6.7	4.7.2.7
Subgroup V (All lots with pointer tactiles)		
Torque and bending moment (at room ambient)	3.5 and 3.6	4.7.2
Insert	3.6.2	4.7.2.2
Tactile caps	3.6.6	4.7.2.6
Subgroup VI (All lots with round concentric types)		
Torque (at room ambient)	3.5 and 3.6	4.7.2
Insert	3.6.2	4.7.2.2
Caps	3.6.5	4.7.2.5
Tensile (at room ambient)	3.5 and 3.6	4.7.2
Dial skirts or plates	3.6.7	4.7.2.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

4.6.1.4.1 Sampling plan. -The sampling plan shall be in accordance with Standard MIL-STD-105 for small-sample inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The AQL shall be 6.5 (percent defective), and the inspection level shall be S-4.

4.6.1.4.2 Disposition of sample units. -Sample units which have been subjected to the Group B inspection may be delivered on the contract or order provided that the inspection lot is accepted and that the knob has not been damaged as a result of conducting the test (must meet 4.6.1.3 Group A inspection).

4.6.1.5 Group C inspection. -Group C inspection shall consist of the tests specified in Table IV in the order shown for each subgroup. Group C inspection shall be made on sample units selected from Group A inspection.

Table IV - Group C inspection

Examination or Test	Requirement Paragraph	Method Paragraph
Subgroup I (3 sample units)		
Sunshine	3.5	4.7.5
Moisture Resistance	3.5	4.7.4
Finish	3.7	4.7.4.1
Subgroup II (3 sample units)		
Salt spray	3.5	4.7.6
Subgroup III (3 sample units)		
Torque (at room ambient)	3.5 and 3.6	4.7.2 and 4.7.3.2
Setscrew (after thermal shock)	3.6.1	4.7.3.2.1
Subgroup IV (3 sample units)		
Thermal shock	3.5	4.7.3 and 4.7.3.1
Insert torque	3.6.2	4.7.3.1.1
Subgroup V (3 sample units)		
Thermal shock	3.5	4.7.3 and 4.7.3.1
Insert torque	3.6.2	4.7.3.1.1
Spinner torque	3.6.3	4.7.3.1.2
Subgroup VI (3 sample units)		
Torque and bending moment (at room ambient)	3.5 and 3.6	4.7.2 and 4.7.3.2
Tactile caps (after thermal shock)	3.6.6	4.7.3.2.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

Table V. Setscrew torque

Setscrew Thread	Torque Reading in Inch Pounds
4-40-UNC-3A	7
8-32-UNC-3A	22

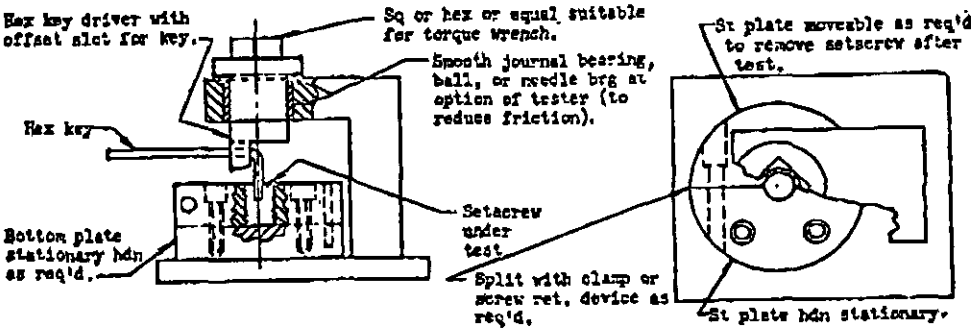


Figure I. Setscrew Test Fixture

4.7.2.2 Insert. - The knobs shall be mounted on a corrosion resistant shaft and held in a suitable jig. The setscrews shall be seated to a minimum of 5 inch pounds for 4-40 and 20 inch pounds for 8-32 not to exceed values in Table V with a suitable torque wrench or driver prior to conducting the test. A torque of 15 inch pounds for 1/8-inch shaft hole and for all No. 4-40 UNC-3A setscrews, and 30 inch pounds for 1/4-inch shaft hole shall be applied to the shaft in a clockwise direction, then in a counter-clockwise direction for a period of 1 minute in each direction. The knobs shall be examined for evidence of movement of the insert within the knob, cracking, or other failure. (See 3.6.2).

4.7.2.3 Spinner and tactile spinner. - Spinner knobs shall be held in a suitable jig to prevent any rotation of the insert. The setscrews shall be seated to a minimum of 5 inch pounds for 4-40 and 20 inch pounds for 8-32 not to exceed values in table V with a suitable torque wrench or driver prior to conducting the test. A 30 inch pound torque shall be applied to the end of the spinner handle perpendicular to the plane of the axes. The torque shall then be applied in a clockwise direction, and then in a counter-clockwise direction for a period of 1 minute in each direction. The spinner knobs shall be examined for evidence of cracking, of movement of the insert or any other member within the knob, or other failure. (See 3.6.3).

4.7.2.4 Spinner-knob balance. - The magnitude of the torque due to imperfect balance of spinner knobs shall be determined by the following method: Use a Waters torque watch gauge or equal. Using a fixture which will rigidly hold the outside case of the gauge allowing the chuck and body of the gauge to rotate freely, with the axis of the gauge in a horizontal position, mount a 1/4-inch shaft, not exceeding 3 inches in length, in the chuck of the gauge. Using the setscrew supplied with the knob, mount the knob on the shaft, with the spinner handle above the shaft at torque dead center (approximately 12 o'clock position). At this point, the dial of the gauge should indicate zero. Rotate the knob slightly from the dead-center position and allow it to find its equilibrium. Observe and record the torque reading of the gauge. Repeat the test with the spinner handle at the lowest point. (See 3.6.4).

4.7.2.5 Caps. - The knobs shall be held in a suitable jig and a radial torque of 20 inch pounds shall be applied to the cap if it has been attached onto the knob body. This test does not apply if the cap is molded integrally with the knob body. (See 3.6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

4.7.2.6 Tactile caps. -The knobs shall be held in a suitable jig and a radial torque of 20 inch pounds shall be applied to the tactile cap (excluding tactile spinners, see 4.7.2.3). Also under the same conditions, the tactile cap shall have a side force of 30 pounds applied to the tactile shape perpendicular to the side near its top surface to provide a bending moment (See 3.6). The knobs shall be examined for evidence of looseness, movement, or separation of the attached part. (See 3.6.6).

4.7.2.7 Dial skirts or plates.- Knobs with dial skirts or plates that are attached to the body of a knob shall be tested by applying a tensile force axially to the dial skirt or plate in accordance with value in table VI. The knobs shall be examined for evidence of looseness, movement, or separation of the attached part. (See 3.6.7).

Table VI. Tensile test for dial skirts or plates

Sizes of Knobs	Tensile force in pounds
up to 3/4" dia.	20
3/4" to 1 1/2" dia.	30
1 1/2" to 2 1/2/ dia.	50

4.7.3 Thermal shock.- Unless otherwise specified (see 3.1 and 4.7.3.2), knobs shall be conditioned by being subjected to method 107, test condition B of Standard MIL-STD-202. At the conclusion of this conditioning, the knobs shall be examined for visible deterioration. (See 3.5 and 4.7.3.1).

4.7.3.1 Tests performed during thermal shock.- The torque tests specified in 4.7.3.1.1 and 4.7.3.1.2 shall be performed during the fifth (final) cycle of the thermal shock test at the conclusion of steps 1 and 3 before returning to room temperature (see tables II and IV). At the conclusion of these tests after specimens return to room temperature, the tests specified in 4.7.3.2 shall be performed (see 3.5, 4.4, and Tables II and III).

4.7.3.1.1 Insert.- While setscrews are seated and knobs are held in the jig, a torque of 15 inch pounds for 1/8-inch shaft hole and for all No. 4-40 UNC-3A setscrews, and 30 inch pounds for 1/4-inch shaft hole shall be applied to the shaft in a clockwise direction, then in a counter-clockwise direction for a period of 1 minute in each direction. The knobs shall be examined for evidence of movement of the insert within the knob, cracking, or other failure. (See 3.6.2).

4.7.3.1.2 Spinner and tactile spinner.- While setscrews are seated and knobs are held in the jig, a torque of 30 inch-pounds shall be applied to the end of the spinner handle perpendicualr to the plane of the two axes. The torque shall then be applied in a clockwise direction, then in a counter-clockwise direction for a period of 1 minute in each direction. The knobs shall be examined for evidence of movement of the insert within the knob, cracking, or other failure. (See 3.6.2).

4.7.3.2 Tests performed after thermal shock.- At the conclusion of thermal shock tests (see 4.7.3) and sample units have returned to room temperature under inspection conditions of 4.4, the tests specified in 4.7.3.2.1 through 4.7.3.2.4 shall be performed. (See Tables II and III, and 3.6).

4.7.3.2.1 Setscrews.- Setscrew shall be removed from sample units with the torque wrench or driver and tested as in 4.7.2.1 (See 3.6.1).

4.7.3.2.2 Caps.- Caps shall be tested as in 4.7.2.5 (See 3.6.5).

4.7.3.2.3 Tactile caps.- Tactile caps shall be tested as in 4.7.2.6 (See 3.6.6).

4.7.3.2.4 Dial skirts or plates.- Dial skirts or plates shall be tested as in 4.7.2.7 (See 3.6.7).

4.7.4 Moisture resistance.- Knobs shall be conditioned by being subjected to method 106 of Standard MIL-STD-202. Initial measurements, polarization, loading voltage, and final measurement are not applicable. At the conclusion of this conditioning, the knobs shall be examined for visible deterioration and tested for finish as specified in 4.7.4.1 (See 3.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

4.7.4.1 Finish.- After completion of moisture resistance conditioning as specified in 4.7.4, the knobs shall be wiped dry, and within 1 minute several lines shall be scratch to a depth of at least 0.003 inch with a sharp ponted stylus. Two parallel lines at least 1/4-inch long and 1/8-inch apart, and two other equal parallel lines intersecting the first two at right angles, shall be scratched on several smooth surfaces (cap, skirt, dial skirt, when available). A single line shall also be scratched on the crest of two or more serrations, parallel to the grooves. A 1/2-inch wide strip of tape conforming to PP-T-60 of sufficient length to cover the lines shall be firmly pressed with the adhesive side against the scratched lines. The tape shall be abruptly pulled off in one motion. The surface of the knobs shall then be examined. (See 3.7).

4.7.5 Sunshine.- Test knobs shall be conditioned by being subjected to method 505 of Standard MIL-STD-810. At the conclusion of this conditioning, the knobs shall be examined for visible deterioration. Corrosion resistant shaft mounting optional for this test. (See 3.5).

4.7.6 Salt-spray (corrosion).- Knobs shall be conditioned by being subjected to method 101, test condition B, of Standard MIL-STD-202. At the conclusion of this conditioning, the knobs shall be lightly washed and dried. They shall then be examined for visible deterioration. (See 3.5).

4.7.7 Flammability.- This test shall be performed on one sample unit of each knob form and color, and one tactile form represented by the sample units. The knob shall be tested as specified in method 111 of MIL-STD-202. The following detail instructions apply to this specification: the knob shall be mounted on a horizontal shaft located on the center of a 1-foot square, vertically mounted panel, with the back surface of the knob not less than 1/8 nor more than 1/4 inch from the surface of the panel. The axis of the flame shall be in the plane of the axis of the shaft and at an angle of 45 degrees below the axis of the shaft. The flame shall be applied to the lowest portion of the knob at the junction of the knob body with the cap or top surface of the knob. Pointer and bar knobs shall be positioned with the pointer pointing upward. (See 3.8).

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Control knobs shall be preserved and packaged Level A or C, packed Level A, B, or C and marked in accordance with MIL-E-17555 as specified (see 6.1).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable MS military standard, and the MS part number. (See 3.1).
(c) Selection of applicable levels of packaging and packing (see 5.1).

6.2 Definitions.- The following definitions are applicable to this specification:

(a) Body	-	That part of the knob containing the gripping surface, and if non-metallic, containing the insert.
(b) Cap	-	The top portion of the surface of the knob may be integral or attached separately.
(c) Cored	-	As used in this specification, to signify the hollowed out portion of the inside or underside of the knob or device.
(d) Crazing	-	Minute cracks on the surface of the knob.
(e) Design	-	Those factors affecting the aesthetic appearance of the knob such as configuration of the ribs, shape of the cap, shape of the dial, etc.
(f) Dial plate	-	Flat dial round or configured as an indicator attached to the bottom of the body visually containing calibration mark or number and also used to conceal the panel locking nut.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

(g) Dial skirt - Truncated cone attached to the bottom of the body visually containing calibration mark or numerals and also used to conceal the panel locking nut.

(h) Finish - The surface or skin of the knob which imparts the lusterless effect.

(i) Fluorescent - As used in this specification to be distinguished from phosphorescent, a marking material which is visible in ordinary light but only visible by the use of black light (radiant energy) in darkness.

(j) Form - The specific form of the knob i. e. - round, pointer, plain, ring skirt, dial skirt, spinner knob, or tactile cap.

(k) Key - A wrench of hexagonal cross section bent at right angles with one leg shorter than the other used to turn socket setscrews.

(l) Marking - As used in this specification in the various sections, it is used to mean
1. Calibration marking as an arrow, line, dot.
2. Identification marking for manufacturer's name or code letter and MS number.
3. Identification marking on packaging.

(m) Phosphorescent - As used in this specification to be distinguished from fluorescent, a marking material which is visible in daylight and absorbs radiant energy and continues to emit energy to make it visible in darkness. Also visible in darkness under radiant energy.

(n) Ream - As used in this specification, it implies rounding off the corners on a hexagon key or hexagon socket setscrew.

(o) Sample units - Those specific components which are undergoing examination, test, or environmental conditioning as opposed to a "sample" which is a random group of components removed from a lot for AQL inspection purposes.

(p) Skirt - Round spacer attached to part of the knob body to conceal the panel locking nut.

(q) Socket - As used in this specification, the hexagon shaped hole on the top end of the setscrew.

(r) Spalling - Splits or splinters on the plastic surface of the knob.

(s) Spinner - A crank attached to the cap portion of the knob body so that continuous rotation may be imparted to the knob. Sometimes synonomous with "crank knob".

(t) Spinner tactile - Same as Spinner except that the crank handle may be of a tactile shape or color.

(u) Strip - As used in this specification there are two meanings. One is synonomous with "Ream" - rounding of the corners of key or hexagon socket of a setscrew; the other is the threads of the setscrew or the tapped hole in the knob body.

(v) Tactile - A strategic and identifying shape both in form and color coding so that visual touch (braille), identification can be made of the function of the shape and a visual identification may be made of the function by color.

(w) Tactile cap - A cap which has both tactile shape and color attached to the knob body.

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 3926 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.

offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Electronic Systems Command Department of the Navy, Washington, D.C. 20360 and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4).

6.4 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Custodians:
Army - EL,
Navy - EC
Air Force - 11
Review Activities:
Army - EL, AT, AV, ME, MI, and MC
Navy - SH
Air Force -
User Activities:
Army - MU
Navy - CG, MC, OS
Air Force -

Preparing Activities:
Navy - EC
(Project 5355-0029)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:12Z
Check the source to verify that this is the current version before use.